

Increased trunk stability modulates sagittal plane lower extremity biomechanics during single-leg landings

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BACKGROUND

- Core stability training is often included in injury prevention program (Steffen et al. 2008, Willson et al. 2005, Hewett et al. 2005).
- Some studies have shown that core stability training modify functional performances (Stanton 2004, Tse et al. 2005, Mills et al. 2005, Butcher 2007, Lust et al. 2009, Sato and Mokha, 2009, Kahile and Gribble 2009, Filipa et al. 2010, Hessari 2011, Saeterbakken 2011, Parkhouse and Ball 2011).
- However, little is know about how increased core stability may modify lower extremity biomechanics and influence lower extremity injury risk

PURPOSES

- To examine how increased trunk stability modulates sagittal plane lower extremity biomechanics during single-leg landings

METHODS

Subjects

- 18 females were recruited for the study
 - 20.2±1.2 yrs, 52.7±7.1 kg, 1.58±0.06 m
 - Exercise at least 30 min / day, 3 times / a week
 - No history of lower extremity surgery
 - No current pain in their low back or extremities

Protocol

- Single-leg drop landing from 30 cm box in three ways

① Self-selected landing (SSL)



- Stood on a dominant leg (preferred leg for landing)
- Arms crossed on the chest
- Landed with the dominant leg in their preferred ways
- Balanced for 3 sec after landing

② Valsalva landing (VLL)



- Mouthpiece with a small hole in their mouth
- Forcefully exhaling just before dropping off the box to the end of landing while hollowing their lower abdomen
- Performed single-leg landing as SSL

③ Enhanced Trunk Stability Landings (TSL)



- A straight cardboard bar attached externally from their sacrum to thorax using an elastic bandage
- Performed VLL

METHODS

④ Practice with Biofeedback



- Performed before both VLL and TSL
- Muscle activations for the anterior abdominnal muscles were monitored
- Exhaled forcefully while hollowing their lower abdominal while having the mouthpiece
- Had them try to activate TrA
- Continued until subjects successfully activated TrA

- SSL performed always first, order of VLL and TSL counterbalanced

Data Reduction and Statistical Analyses

- Analyzed from the Foot Contact to 1) Peak Knee Extensor Moment, 2) peak vertical GRF, and 3) when body COM reached the lowest position
- 2-Way Repeated Measure ANOVA for Thigh-Sacrum and Thigh-Thorax Flexion Angles
- 1-Way Repeated Measure ANOVA for Kinetic Variables
- Post-hoc Bonferroni corrections comparing landing conditions

RESULTS

Fig. 1 Normalized Lower Extremity Peak Extensor (Plantar Flexor) Moments And Vertical GRF During Single-leg Landing

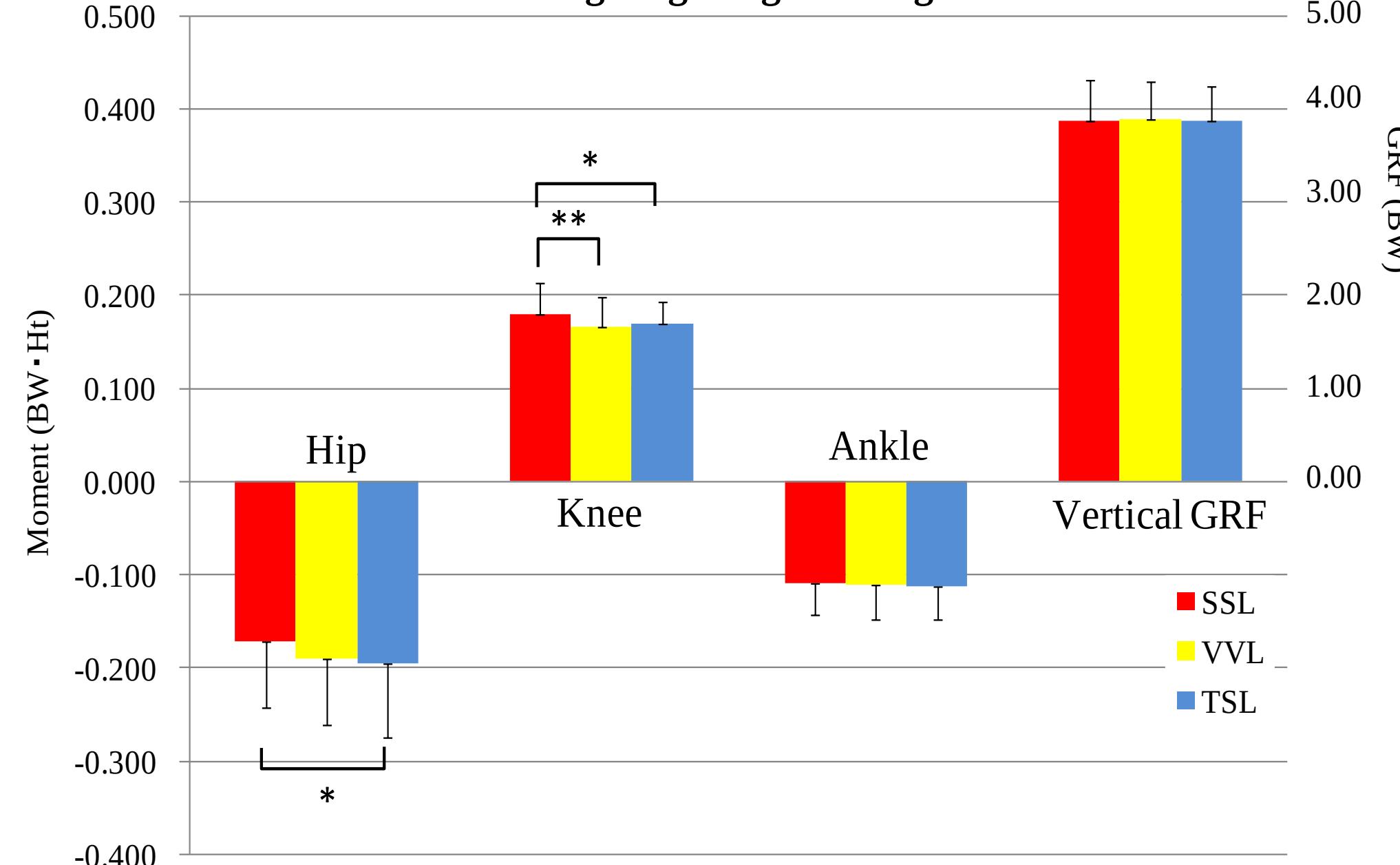


Fig. 2 Thigh-Sacrum and Thigh-Thorax Flexion Excursions From The Foot Contact To Peak Vertical GRF

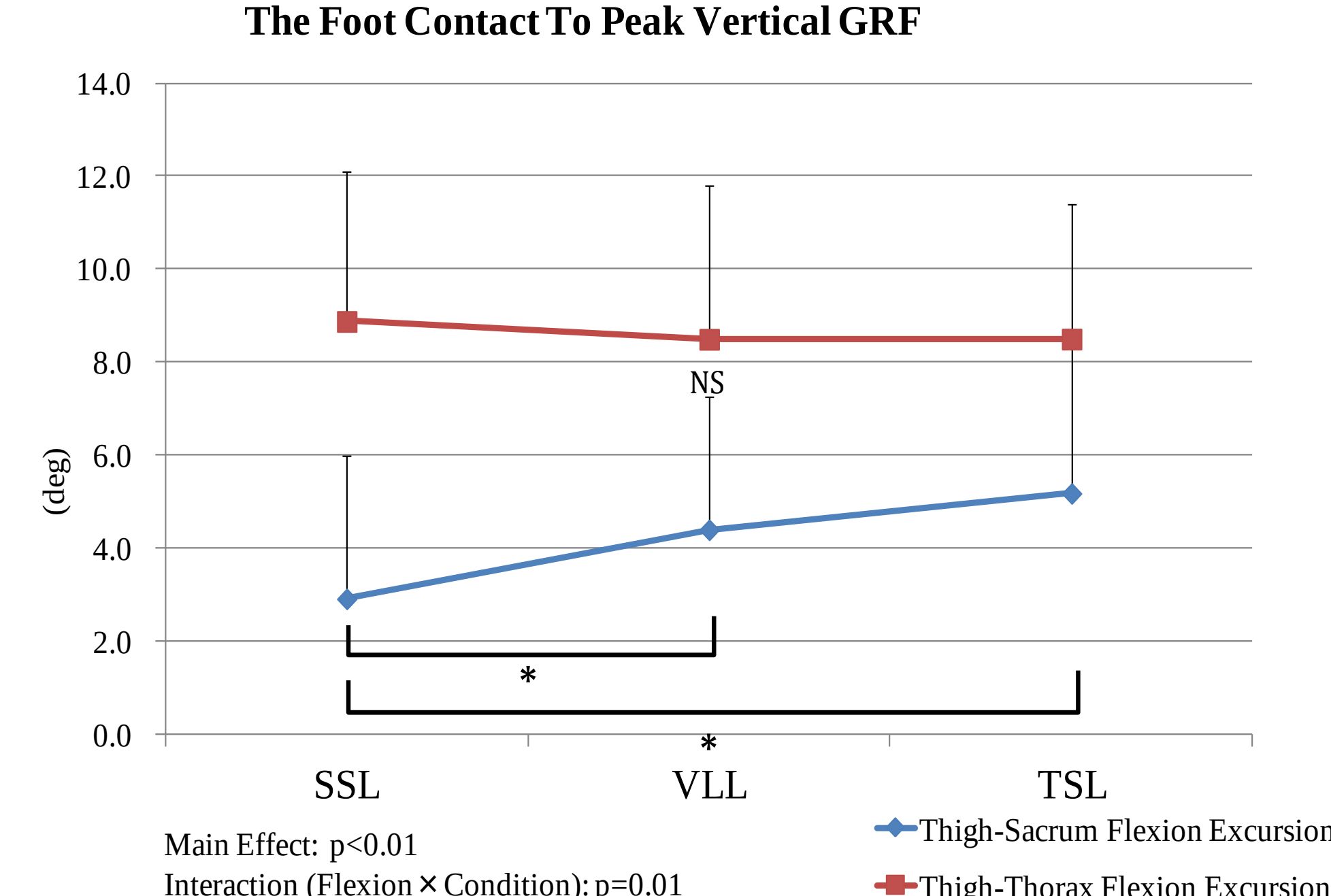
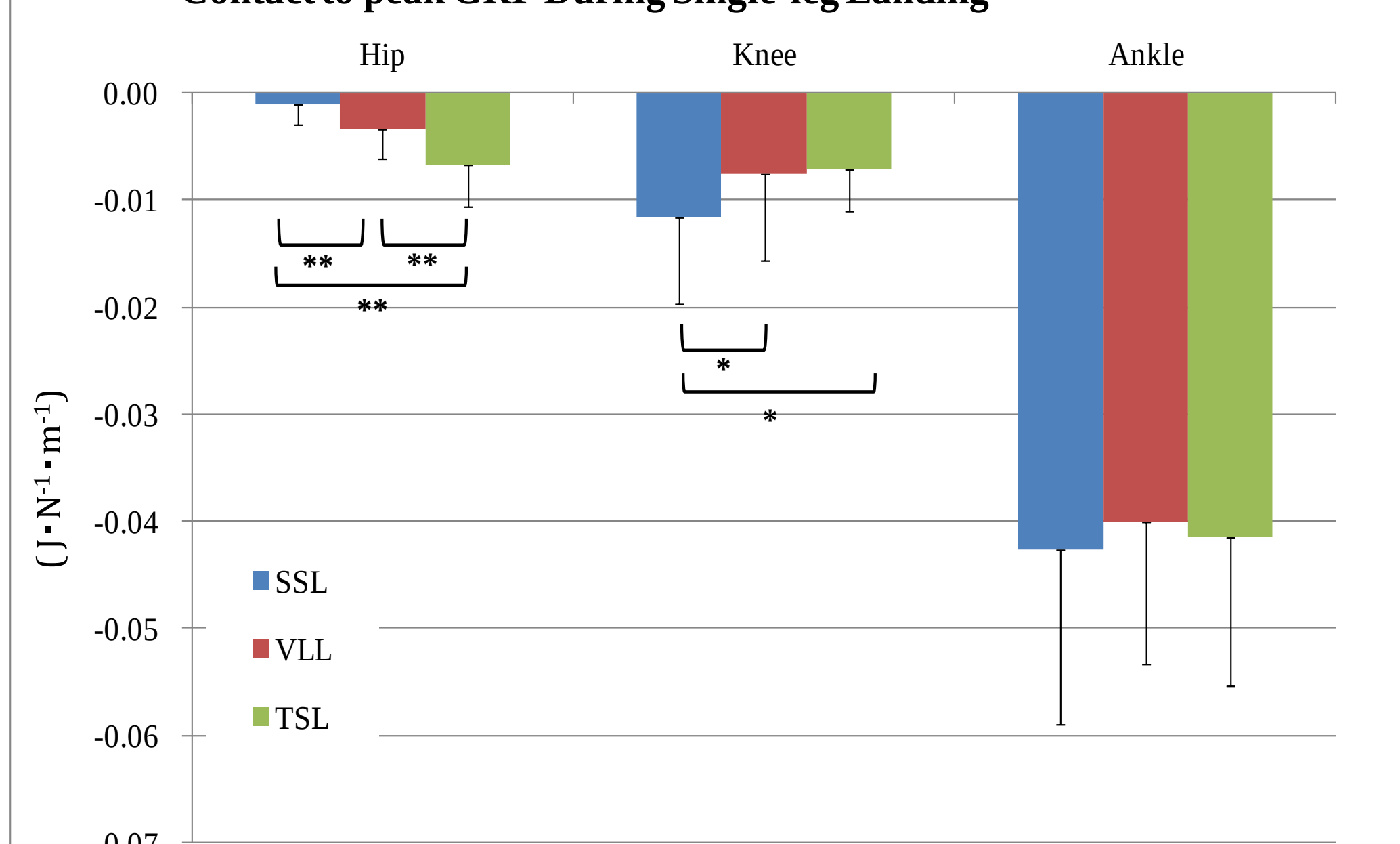


Fig. 3 Lower Extremity Sagittal Plane Negative Joint Work From the Foot Contact to peak GRF During Single-leg Landing



RESULTS

Fig. 4 Thigh-Sacrum and Thigh-Thorax Flexion Excursions From The Foot Contact To Peak Knee Extensor Moment

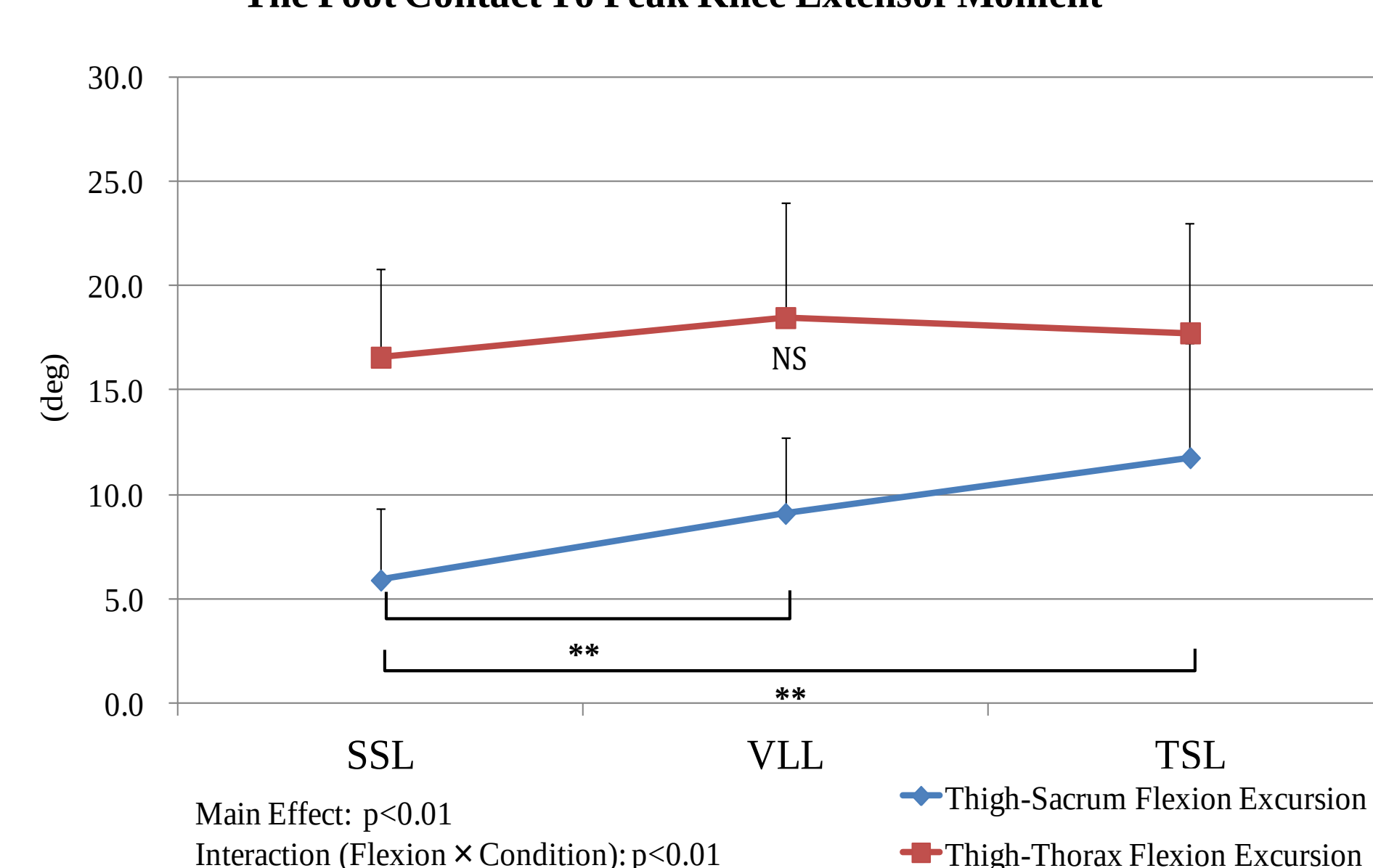


Fig. 5 Lower Extremity Sagittal Plane Negative Joint Work From the Foot Contact to peak Knee Extensor Moment During Single-leg Landing

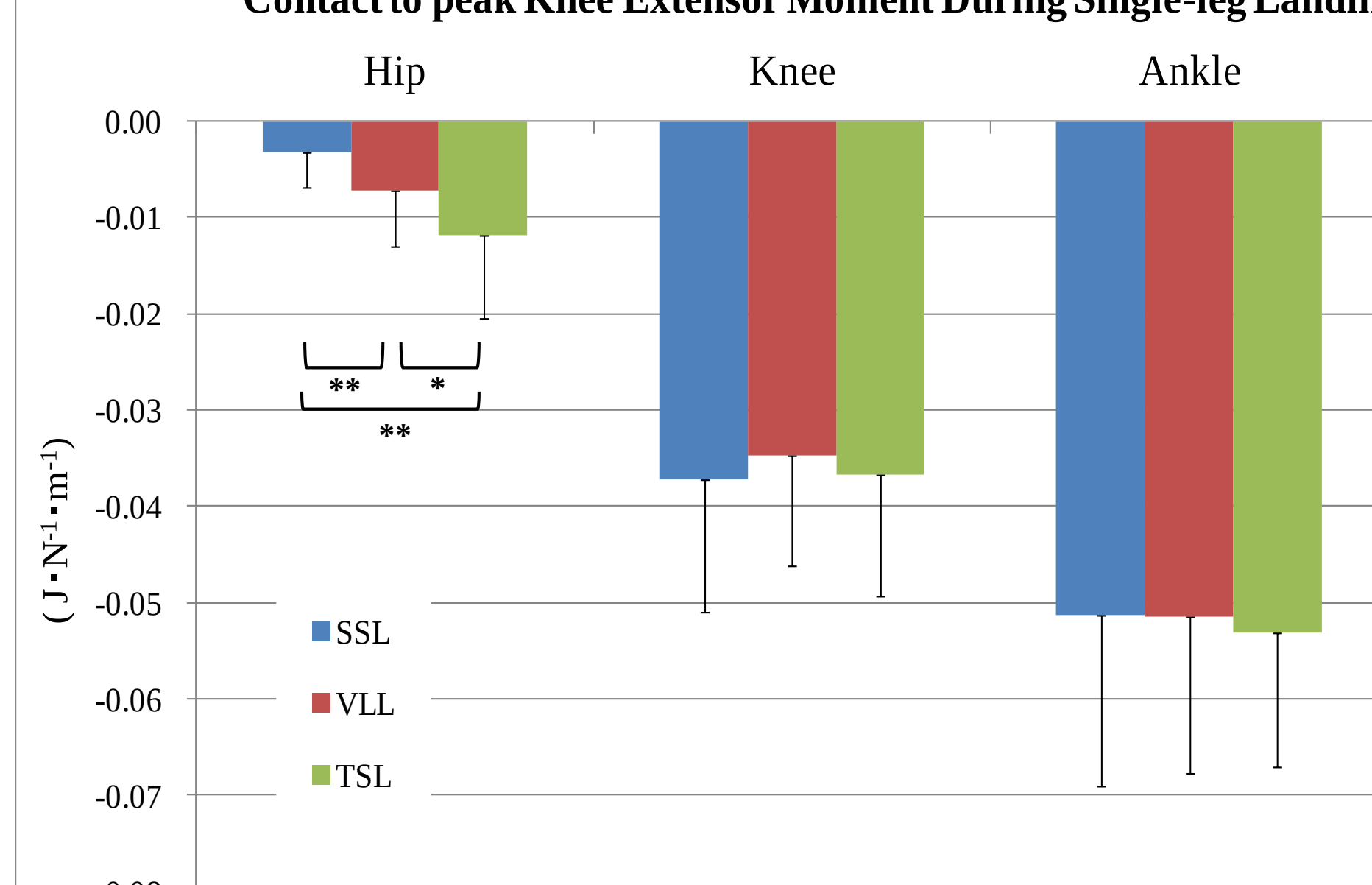


Fig. 6 Thigh-Sacrum and Thigh-Thorax Total Flexion Excursion During The Descending Phase of Single-leg Landings

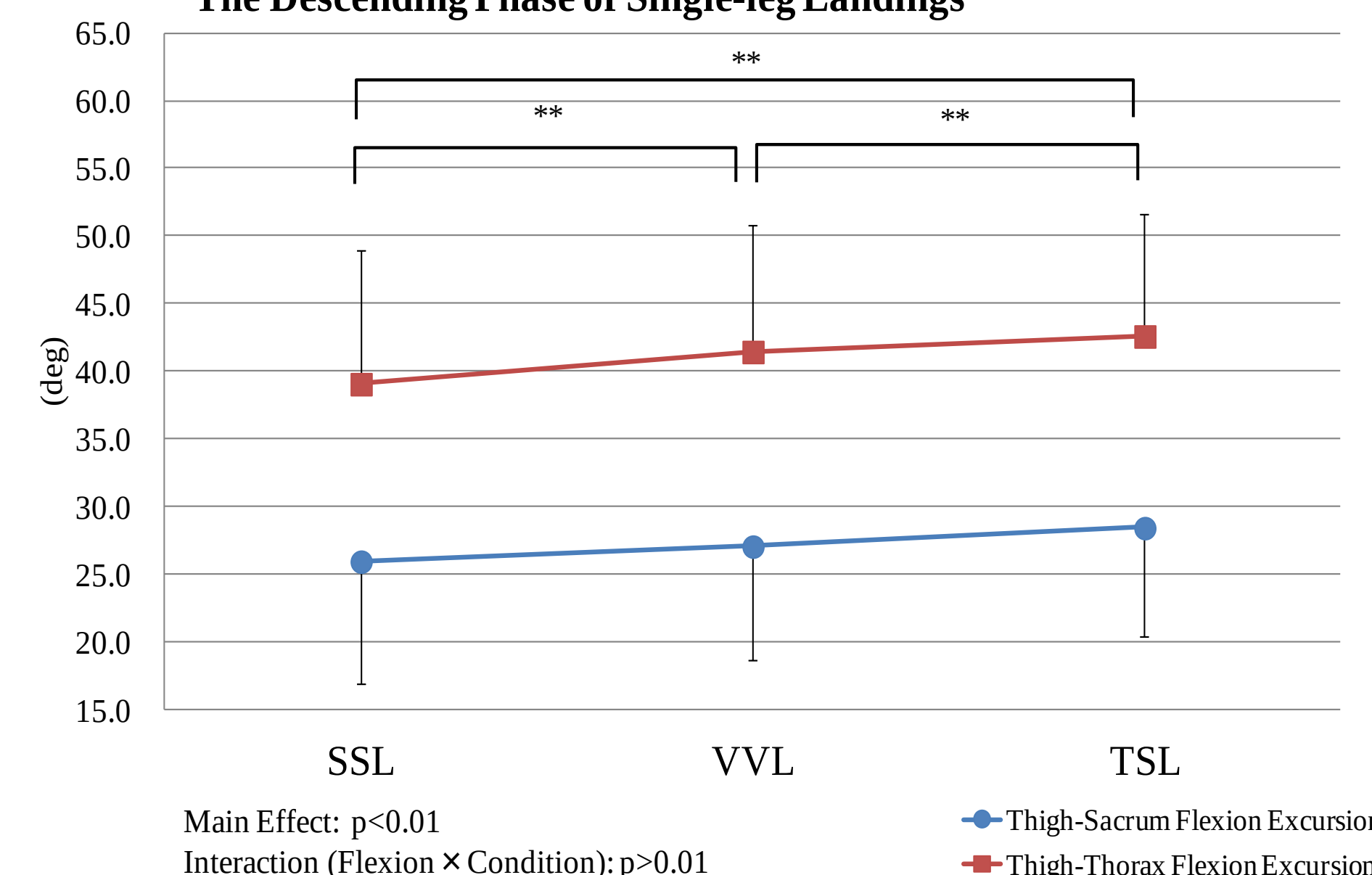
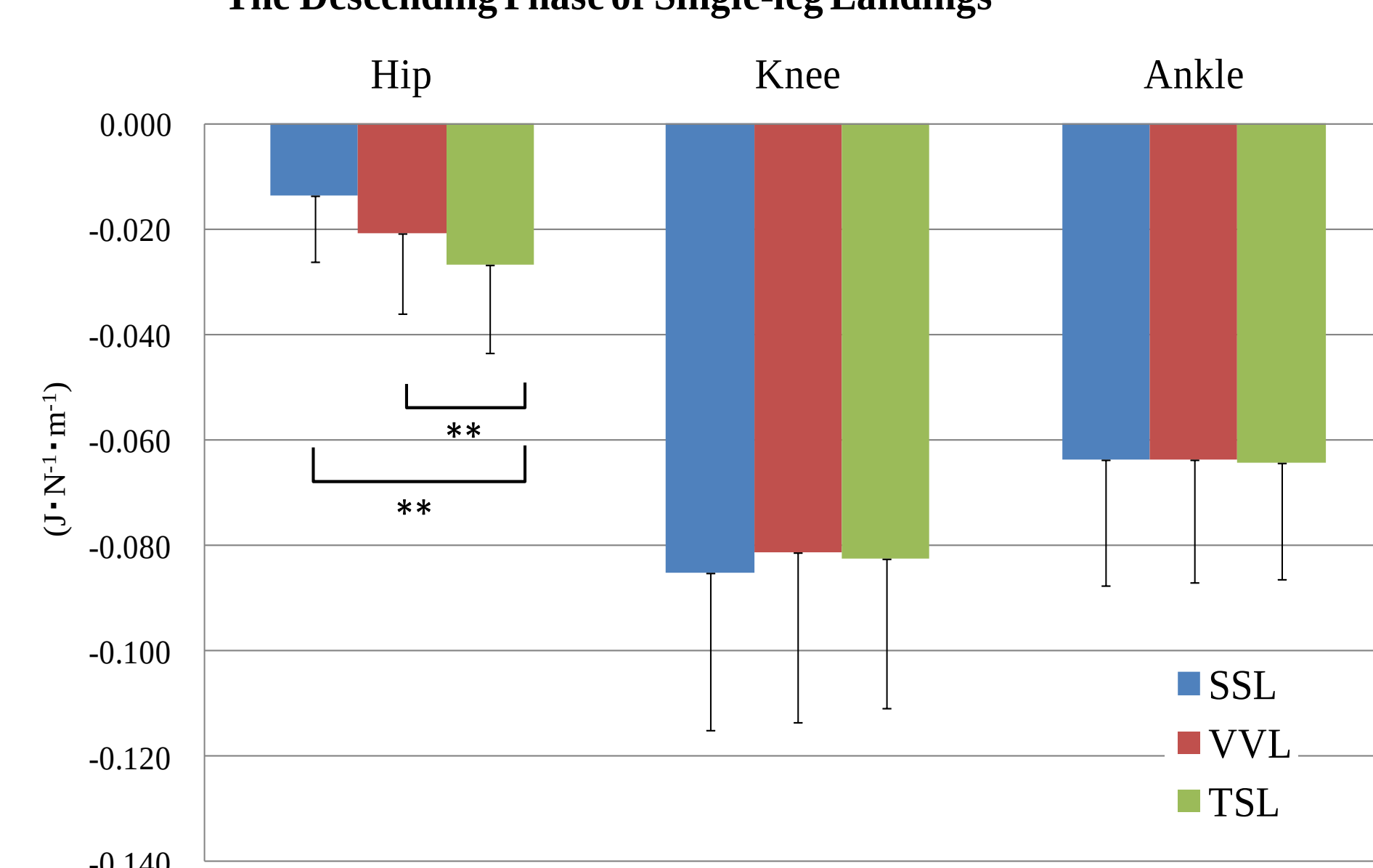
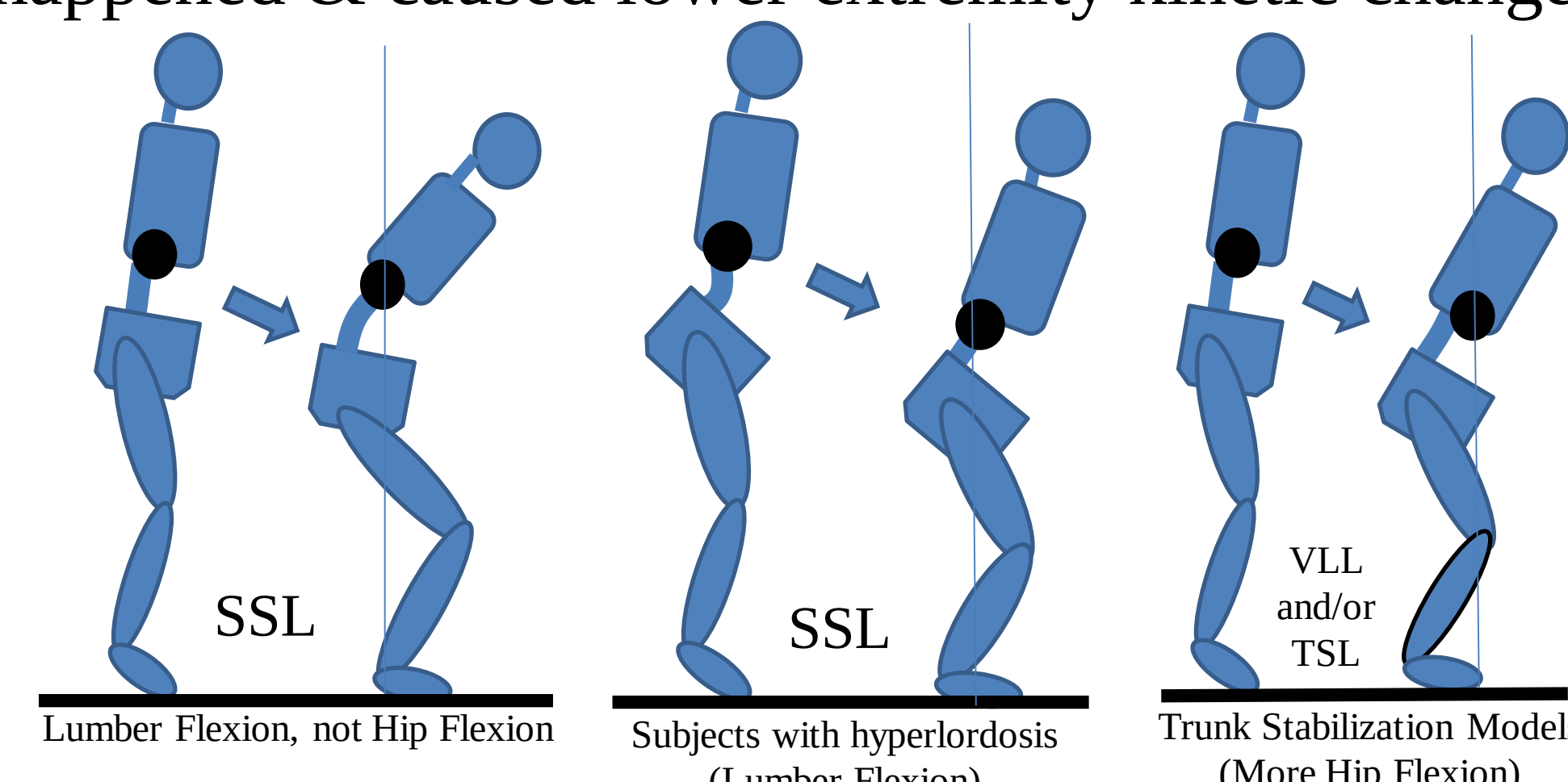


Fig. 7 Lower Extremity Sagittal Plane Total Negative Joint Work During The Descending Phase of Single-leg Landings



DISCUSSIONS & CONCLUSIONS

- The following postural modifications may have happened & caused lower extremity kinetic changes



- ↑ed hip joint flexion ROM, hip extensor moment production demand and ↓ed knee extensor moment production demand
- ↑ed hip extensor contraction forces due to ↑ed hip extensor muscle length
- These lower extremity kinetic changes may influence knee stability and injury risks

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